

Agentic RFP Evaluation and Supplier Ranking

Prepared by kvl.narasimharao

This report presents a working application that compares supplier responses to a request for proposal. It explains how proposal PDFs are read, how criterion scores are checked, how Python calculates peer results and final ranks, and what the four-supplier demonstration produced. The repository includes the running application, sample proposals, saved results, tests, and screenshots.

Project purpose

A procurement team must review several supplier proposals against the same requirements. Manual reading makes the comparison slow and can lead to inconsistent scoring. This application produces an evidence-linked scorecard for each proposal and applies one published set of arithmetic and ranking rules to every supplier.

The language model judges proposal content and supplies criterion scores, reasons, and evidence. Python validates that response, calculates every comparison measure, and assigns the final order. The model does not set benchmarks or ranks.

What the application does

- Loads active criteria and weights from SQLite and displays them in the Criteria tab.
- Accepts at least two supplier PDFs with supplier name, submission date, and historical experience rating.
- Evaluates each PDF independently against the criteria active at the start of the batch.
- Corrects missing, malformed, duplicate, and out-of-range model results and records each correction.
- Calculates weighted scores, peer benchmarks, gaps, relative percentages, PPI, and final ranks in Python.
- Saves the full batch under one run ID and offers leaderboard, scorecards, and JSON download.

Submission requirements and included files

The source brief is Agentic_RFP_Evaluation_Mini_Project.pdf, section 10, Submission Requirements. Every requested item is present in the public repository or at the live application link.

Required item	Included deliverable
Complete source code	streamlit_app.py, rfp/, scripts/, tests/, and requirements.txt in the public repository
SQLite setup	scripts/seed_db.py and rfp/database.py create and seed five sample criteria
Four supplier PDFs	data/proposals/apex_systems.pdf, brightpath_tech.pdf, nexaworks.pdf, orbit_digital.pdf
Public deployed app	https://rfp-evaluation-kvlnarasimharao.streamlit.app/
README	README.md contains setup, architecture, formulas, assumptions, and screenshots
Completed JSON export	data/live_run.json and data/hosted_run.json store completed four-supplier runs
Short demonstration	docs/Demonstration.md shows a completed run and a validation case with screenshots

Source code structure

Path	Responsibility
streamlit_app.py	Upload form, criteria editor, leaderboard, detailed scorecards, saved runs, export
rfp/evaluation.py	PDF extraction, dynamic prompt, model request, response normalization
rfp/scoring.py	Weighted arithmetic, peer comparison, deterministic ranking
rfp/workflow.py	Ordered batch evaluation and persistence
rfp/database.py	SQLite schema, criteria operations, completed run records
scripts/ and tests/	Sample generation, database seed, live run, document build, automated checks

System workflow

The user uploads the proposals and enters metadata. The orchestrator reloads active criteria, reads text from each PDF, and builds a prompt from those database rows. Gemini or Ollama Cloud returns a JSON assessment for one supplier at a time. The validator creates a complete, bounded scorecard even when fields are missing or malformed. Once all suppliers have scorecards, Python calculates the peer measures and sorts the batch. The complete result is committed to SQLite and shown in Streamlit.

Stage	Input	Output
Setup	SQLite criteria	Active criteria with total weight of 100%
Evaluate	PDF text and active criteria	Scores, reasons, evidence, risks, summary
Validate	Model JSON	Complete scorecards and correction records
Compare	Validated supplier scores	Benchmarks, gaps, relative results, PPI
Finish	Peer results and metadata	Ranks, saved run, leaderboard, JSON export

Evaluation criteria

Criteria are stored in SQLite rather than fixed in the prompt. The editor allows activation, deactivation, and weight changes. A batch cannot start unless active weights total 100%. Each criterion uses a maximum score of 10 in the included demonstration.

Criterion	Weight	What is inspected
Technical Capability	30%	Architecture, integrations, scalability, technical fit
Implementation Plan	20%	Timeline, milestones, staffing, risk plan
Commercial Value	20%	Pricing clarity, total cost, assumptions
Security & Compliance	20%	Controls, certifications, privacy, auditability
Support & Experience	10%	Support model, similar projects, references

Scoring and tie breaks

For each criterion, the absolute contribution is the validated score divided by its maximum score, multiplied by the criterion weight. The absolute score is the sum of those contributions. The benchmark is the highest validated score for that criterion among suppliers in the batch. A supplier's gap is its score minus the benchmark, so a benchmark leader has a gap of zero.

Relative performance is supplier score divided by benchmark, multiplied by 100. When every supplier scores zero on a criterion, the benchmark is zero and relative performance is set to 100% for all suppliers on that criterion. PPI is the sum of each relative percentage multiplied by its criterion weight divided by 100. It is calculated before display rounding is applied.

Final sorting is fixed: higher PPI first, then earlier submission date, then higher historical experience rating, then supplier name in ascending alphabetical order. Sequential ranks start at 1 after sorting. Experience affects a tie break only; it does not change criterion or absolute scores.

Worked calculation for Apex Systems

Apex received scores of 9, 9, 7, 9, and 8 out of 10 for the five criteria. Its absolute score is $(9/10 \times 30) + (9/10 \times 20) + (7/10 \times 20) + (9/10 \times 20) + (8/10 \times 10) = 85$. The benchmark for each criterion in the four-supplier run was 9. Apex therefore had relative results of 100%, 100%, 77.7778%, 100%, and 88.8889%. Applying the five weights gives a PPI of 94.4445 after display rounding.

Model response validation

The evaluator requests valid JSON with one entry per active criterion, including criterion ID, score, explanation, and supporting evidence. The validator uses the database criterion IDs and maximum scores as the source of truth. It applies the following rules before any arithmetic:

- Invalid JSON or a non-object response becomes an empty assessment; missing criteria receive zero.
- Unknown and duplicate criterion IDs are ignored. Missing expected criteria receive zero.
- Non-numeric or non-finite scores become zero. Scores outside the range are clipped.
- Missing explanations receive a placeholder. Missing evidence is recorded for review.
- A malformed risks field becomes an empty list. Every correction is recorded with the supplier name.

Supplier names must be unique within a batch. Experience ratings must be between 0 and 10. Submission dates are checked before ranking. The application rejects PDFs that contain too little selectable text.

Stored data and output

SQLite has three tables: evaluation_criteria, rfp_runs, and supplier_results. The first stores the editable rubric. The run table stores the creation time, criteria snapshot, correction records, and complete result JSON. The supplier table stores each supplier's metadata, absolute score, PPI, rank, and scorecard. One RFP_RUN_ID connects all suppliers in a batch. The API key is not written to these tables or to the JSON export.

The interface has Evaluate, Criteria, Saved runs, and Sample run tabs. A completed run displays the ranked leaderboard, every criterion's score and evidence, peer comparisons, validation corrections, and a JSON download. The Sample run tab includes the completed Gemini result and a separate validation example.

Demonstration data

Supplier	PDF file	Intended difference
Apex Systems	apex_systems.pdf	Strong design and security; higher price
BrightPath Tech	brightpath_tech.pdf	Lowest price and fast timeline; weak compliance detail
NexaWorks	nexaworks.pdf	Balanced cost; strongest plan and support
Orbit Digital	orbit_digital.pdf	Strong references; vague integration scope

Four fictional two-page proposals were prepared for the same university procurement request. They differ in architecture, implementation timing, cost, security detail, support, and references. Gemini 2.5 Flash evaluated all four in a completed batch. The table reports Python's final calculations; scores in the five criterion columns are out of 10.

Rank	Supplier	Tech	Plan	Value	Security	Support	Absolute	PPI
1	Apex Systems	9	9	7	9	8	85.0	94.4445
2	NexaWorks	9	9	8	6	9	82.0	91.1111
3	Orbit Digital	6	7	6	7	9	67.0	74.4445
4	BrightPath Tech	4	6	9	1	3	47.0	52.2222

Apex Systems ranked first with strong technical and security detail. NexaWorks followed with a strong plan and support offer. Orbit Digital had substantial references but less detail on ERP integration. BrightPath Tech had the lowest price but limited security detail. The model returned a malformed risks field for BrightPath; validation replaced it with an empty list and retained its valid criterion scores.

The saved run ID is d3bee8e3-cb68-4183-8d1a-9d9429742653. The complete result is in data/live_run.json. A separate data/sample_run.json deliberately includes an out-of-range score so the correction path can be demonstrated without a model call.

Testing and verification

Check	Result
Response normalization	Missing results, duplicate IDs, invalid values, and clipping are covered.
Scoring and ranking	Weighted score, peer measures, zero benchmark, and every tie break are covered.
Persistence	A complete two-supplier run is saved and loaded from SQLite.
Application	The live app opens; Sample run shows four ranks, scorecards, corrections, and JSON download.

All four automated tests passed. The four proposal PDFs were evaluated in one Gemini batch. A separate test uploaded the revised PDFs to the public Streamlit app, completed and saved a new four-supplier run, then downloaded its JSON export. The public app also displayed the criteria, detailed scorecards, and validation example. Streamlit application testing loaded all four tabs with no exceptions.

Demonstration sequence

Successful run: open the live app and select Sample run, then Live Gemini evaluation. The leaderboard shows four final ranks. Expand Apex Systems to inspect the criterion explanations, supporting evidence, benchmark, gap, and relative percentages. Use Download complete result as JSON to export the saved run.

Validation case: in Sample run, select Validation example. Its deliberately out-of-range criterion score is clipped to zero before scoring, and the correction is recorded in Run details. The completed Gemini example also corrects a malformed risks field for BrightPath Tech. These cases demonstrate that model output is checked before ranking.

How to run and present the project

Public repository: <https://github.com/kvlnarasimharao/agentric-rfp-evaluation>

Live application: <https://rfp-evaluation-kvlnarasimharao.streamlit.app/>

In the live app, open Sample run to review the completed comparison, then open Criteria to view the active weights. To evaluate new proposals, use Evaluate, upload at least two text-based PDFs, fill supplier metadata, and select Evaluate suppliers. The hosted Gemini key is held in Streamlit secrets, so no key entry is needed for that deployment.

For a local run, clone the repository, install packages from requirements.txt, run python scripts/seed_db.py, then run streamlit run streamlit_app.py. Set GEMINI_API_KEY locally to evaluate a new batch. The sample run is viewable without a key. Run python -m unittest discover -s tests -v to repeat the automated checks.

Operating conditions

PDFs must contain selectable text; scanned images require text recognition before upload. The model's judgment and cited evidence should be reviewed for procurement use. The public cloud deployment uses local SQLite storage that may reset when the app instance restarts, so completed results should be downloaded as JSON when a durable record is needed.

References

Source brief, Agentic_RFP_Evaluation_Mini_Project.pdf, sections 1 to 10

Google AI for Developers, Gemini generateContent API: <https://ai.google.dev/api/generate-content>

Streamlit, Deploy your app on Community Cloud: <https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app/deploy>

Streamlit, Secrets management: <https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app/secrets-management>

Application screenshots

Completed four-supplier evaluation in the Sample run tab.

RFP Supplier Evaluation

Compare supplier proposals using document evidence and transparent ranking rules.

Evaluate Criteria Saved runs Sample run

Example

Live Gemini evaluation

Both examples use the four fictional supplier proposals.

Leaderboard

Rank	Supplier	Absolute score / 100	PPI / 100	Submitted	Experience / 10
1	Apex Systems	85	94.4445	2026-09-15	8.5
2	NexaWorks	82	91.1111	2026-09-16	8
3	Orbit Digital	67	74.4445	2026-09-13	9.5
4	BrightPath Tech	47	52.2222	2026-09-14	4

Order: higher PPI, earlier submission, higher experience, then supplier name A to Z.

Active criteria and weights loaded from SQLite.

RFP Supplier Evaluation

Compare supplier proposals using document evidence and transparent ranking rules.

Evaluate Criteria Saved runs Sample run

Evaluation criteria

criterion_id	name	description	weight	max_score	Active
1	Technical Capability	Architecture, integrations, scalability, technical fit	30	10	☒
2	Implementation Plan	Timeline, milestones, staffing, risk plan	20	10	☒
3	Commercial Value	Pricing clarity, total cost, assumptions	20	10	☒
4	Security & Compliance	Controls, certifications, privacy, auditability	20	10	☒
5	Support & Experience	Support model, similar projects, references	10	10	☒

Current active weight: 100%

Save criteria

Hosted evaluation screenshots

The public app accepted the four revised supplier PDFs and their metadata.

New evaluation batch

Upload text-based PDFs. Every file is evaluated with the active criteria shown above.

Supplier proposal PDFs

 apex_systems.pdf 3.5KB

 brightpath_tech.pdf 3.8KB

 nexaworks.pdf 3.9KB

 orbit_digital.pdf 3.9KB

+

Active weights total 100%. Technical Capability 30%, Implementation Plan 20%, Commercial Value 20%, Security & Compliance 20%, Support & Experience 10%

apex_systems.pdf

Supplier name

Submission date

Historical experience / 10

Apex Systems

2026 / 09 / 27

5.00 - +

brightpath_tech.pdf

Supplier name

Submission date

Historical experience / 10

BrightPath Tech

2026 / 09 / 27

5.00 - +

nexaworks.pdf

Supplier name

Submission date

Historical experience / 10

NexaWorks

2026 / 09 / 27

5.00 - +

orbit_digital.pdf

Supplier name

Submission date

Historical experience / 10

Orbit Digital

2026 / 09 / 27

5.00 - +

> AI connection

Evaluate suppliers

The completed batch was saved under one run ID. Its JSON export is included as data/hosted_run.json.

Select a saved run

2026-09-27T11:47:48.812905+00:00 - 5cfb30b2

▼

Leaderboard

Rank	Supplier	Absolute score / 100	PPI / 100	Submitted	Experience / 10
1	Apex Systems	85	94.4444	2026-09-27	5
2	NexaWorks	82	91.1111	2026-09-27	5
3	Orbit Digital	68	75.5556	2026-09-27	5
4	BrightPath Tech	43	47.7778	2026-09-27	5

Order: higher PPI, earlier submission, higher experience, then supplier name A to Z.

Detailed scorecards

> #1 Apex Systems

> #2 NexaWorks

> #3 Orbit Digital

> #4 BrightPath Tech

Run details

5cfb30b2-d822-487d-966c-685c3b452e2f

All scorecards passed normalization without warnings.

Download complete result as JSON

Scorecard and validation screenshots

The scorecard shows each criterion score with its explanation, proposal evidence, benchmark, gap, and relative result.

Detailed scorecards

#1 Apex Systems

Apex Systems proposes a technically sound and scalable web portal solution that meets the university's requirements. The implementation plan is detailed with a clear timeline, milestones, and staffing, fitting within the expected delivery window. Security controls are robust, and the company holds ISO 27001 certification. While pricing is clear, the assumption of a single ERP connector and the potential for additional costs for more connectors or legacy migration are noted risks. Support and relevant experience are demonstrated, though reference contact details are not immediately provided. The proposal addresses most key areas effectively, but the commercial risks related to additional connectors should be carefully considered.

Technical Capability - 9/10 Weight 30% | Best 9 | Gap 0 | Peer 100%

The proposal details a modular architecture, specific integrations (SSO, ERP, PDF, JSON export), and confirms scalability beyond the required user count (200 users, tested 500-user load profile). This directly addresses the technical requirements.

Evidence: Apex proposes a modular web portal with a documented REST API, event-driven ingestion, and a managed relational database. It meets the 200-user target with a tested 500-user load profile. The integration plan names SSO via SAML, procurement ERP exchange through REST, PDF text extraction, audit events, and export to JSON. Architecture diagrams identify separate ingestion, scoring, and review services.

Implementation Plan - 9/10 Weight 20% | Best 9 | Gap 0 | Peer 100%

A clear 15-week timeline is provided, along with specific milestones (discovery, build, integration, testing, handover) and a dedicated project team (project manager, two engineers, security lead, tester). A risk register is mentioned, specifically covering ERP delays, which is a relevant risk.

Evidence: A 15-week plan assigns a project manager, two engineers, a security lead, and a tester. Milestones: discovery weeks 1-2; build weeks 3-9; integration weeks 10-12; testing weeks 13-14; handover week 15. The risk register covers ERP delays.

Commercial Value - 7/10 Weight 20% | Best 9 | Gap -2 | Peer 77.7778%

Pricing is clear for implementation and annual support, with a total first-year cost provided. Key assumptions are stated regarding ERP connectors and legacy file migration, which could impact the total cost if the university has multiple ERPs or needs migration. The separate connector charge is also noted as a risk.

Evidence: Fixed implementation fee: USD 138,000. Annual support: USD 22,000. Total first year: USD 160,000. Assumes one ERP connector and no migration of legacy files; additional connectors are priced separately. The higher first-year price and separate connector charges may increase procurement cost.

Security & Compliance - 9/10 Weight 20% | Best 9 | Gap 0 | Peer 100%

The proposal lists a strong set of security controls, including SSO, least privilege, encryption, quarterly penetration tests, daily backups, and 365-day audit retention. ISO 27001 certification is explicitly stated with a certificate number, demonstrating compliance.

Evidence: Controls include SAML SSO, least-privilege roles, encryption in transit and at rest, quarterly penetration tests, daily backups, 365-day audit retention, and an incident notification target of 24 hours. ISO 27001 certification is stated with certificate number AX-27001-24.

The separate validation example corrects an invalid score before calculating the final order.

Example

Validation example

Both examples use the four fictional supplier proposals.

Leaderboard

Rank	Supplier	Absolute score / 100	PPI / 100	Submitted	Experience / 10
1	NexaWorks	86	92.2222	2026-09-16	8
2	Apex Systems	78	84.3333	2026-09-15	8.5
3	Orbit Digital	67	71.8889	2026-09-13	9.5
4	BrightPath Tech	51	54.6667	2026-09-14	4

Order: higher PPI, earlier submission, higher experience, then supplier name A to Z.

Detailed scorecards

#1 NexaWorks

#2 Apex Systems

#3 Orbit Digital

#4 BrightPath Tech